MATERIAL SCIENCE

25MSC01

ROLL NO : 25CS619

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DEPARTMENT : B.E - CSE(H)

PART-B

01. A student performs Young's Double slit Experiment using monochromatic light of wavelength 600 nm. At a point on the screen, the path difference between two waves is 1200nm. Will the point be bright or dark? why?

The path difference is 1200 nm and wavelength is 600nm.

Path difference = $1200 = 2 \times 600 = 2\lambda$ since, the path difference is an integral multiple of wavelength ($n\lambda$), the point will be bright due to constructive interference.

02. An electron moves with high velocity in a vacuum tube. Why does it show wave-like behaviour, while a moving cricket ball does not?

An electron shows wave-like behaviour due to its very small mass, giving it a noticable de-Broglie wavelength.

A cricket ball has very large mass, so its wavelength is extremly small and cannot be observed. Hence, it does not show wave behavior.

Q3. An electron is observed usig high-energy light to determine its exact position. What happens to the uncertainty in its momentum and why?

Using high-energy light increases the accuracy of position measurement. According to the Heisenberg Uncertainity Principle.

$$\Delta x \cdot \Delta P \geq \frac{h}{4\pi}$$

If uncertainity in position (Δx) decreases, uncertainity in momentum (ΔP) increases. Thus, the electron's momentum becomes more uncertain.

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04. A photon interacts with an excited atom. Under what condition will this interaction produce another identical photon?

An identical photon is produced when the incoming photon causes simulated Emission.

This happens when the atom is in an excited state and the photon energy is excatly equal to the energy difference between two atomic levels.

PART - C

03. Derive the time-independent schrodinger equation and explain the physical significance of the wave function. Apply it to a particle in a one-dimensional potential box and obtain the expression for energy levels.

Derivation:

$\psi \rightarrow$ wave function

Classical differential equation of wave motion is

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} = \frac{1}{v^2} \frac{\partial^2 \psi}{\partial t^2} \Rightarrow (1)$$

$$\nabla^2 \psi = \frac{1}{V^2} \frac{\partial^2 \psi}{\partial t^2} \Rightarrow (2)$$

$\nabla^2 \Rightarrow$ Laplacian operator.

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

The solution of equation (2) is

$$\psi = \psi_0 e^{-i\omega t} \Rightarrow (3)$$

Differentiate eqn (3) with respect to time.

$$\frac{\partial \psi}{\partial t} = -i\omega(\psi_0 e^{-i\omega t})$$

Differentiate with respect to t

$$\frac{\partial^2 \psi}{\partial t^2} = \frac{\partial}{\partial t}\left(\frac{\partial \psi}{\partial t}\right)$$

$$= (-i\omega)(-i\omega)\psi_0 e^{-i\omega t}$$

$$= i^2\omega^2 \psi_0 e^{-i\omega t}$$

$$= (-1)\omega^2 \psi_0 e^{-i\omega t}$$

From eqn (3)

$$= -\omega^2 \psi$$

$$\frac{\partial^2 \psi}{\partial t^2} = -\omega^2 \psi \Rightarrow (4)$$

eqn ④ sub in eqn ②

$$\nabla^2 \psi = \frac{1}{V^2} - \omega^2 \psi$$

$$\nabla^2 \psi = -\frac{1}{V^2} \omega^2 \psi$$

$$\nabla^2 \psi + \frac{\omega^2}{V^2} \psi = 0 \Rightarrow (5)$$

Angular frequency $\omega = 2\pi\nu$

where $\nu = \frac{V}{\lambda}$

$$\omega = 2\pi \frac{V}{\lambda}$$

$$\frac{\omega}{V} = \frac{2\pi}{\lambda} \Rightarrow (6)$$

square on both sides.

$$\frac{\omega^2}{V^2} = \frac{4\pi^2}{\lambda^2} \Rightarrow (7)$$

sub eqn ⑦ in eqn ⑤

$$\nabla^2 \psi + \frac{4\pi^2}{\lambda^2} \psi = 0 \Rightarrow (8)$$

∴ De-Broglie eqn.

$$\lambda = \frac{h}{mv}$$

$$\nabla^2 \psi + \frac{4\pi^2 m^2 v^2}{h^2} \psi = 0 \Rightarrow (9)$$

Total Energy of Particle.

$$E = P.E + K.E$$

$$E = V + \frac{1}{2} mv^2$$

$$mv^2 = (E - V) 2$$

multiple by m on both sides.

$$m^2 v^2 = 2m (E - V) \Rightarrow (10)$$

sub eqn (10) in eqn (9)

$$\nabla^2 \psi + \frac{4\pi^2}{h^2} 2m (E - V) \psi = 0 \Rightarrow (11)$$

$$\nabla^2 \psi + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

reduced plank's constant.

$$\hbar = \frac{h}{2\pi}$$

$$\hbar^2 = \frac{h^2}{4\pi^2}$$

$$\frac{1}{\hbar^2} = \frac{4\pi^2}{h^2} \Rightarrow (12)$$

eqn (12) sub in eqn (11)

$$\boxed{\nabla^2 \psi + \frac{1}{\hbar^2} 2m (E - V) \psi = 0}$$

∴ Schrodinger Time Independent wave equation.

Physical significance of wave equation. Function.

ψ = wave function.

$|\psi|^2$ gives probability density.

Must be continuous and finite.

Particle in One-Dimensional Box

conditions:

particle confined betwen $x=0$ and $x=L$

$V=0$ inside, infinite outside

soln:

$$\psi_n = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}$$

Energy levels.

$$E_n = \frac{n^2 h^2}{8mL^2}, \quad n = 1, 2, 3 \ldots$$

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04. Lasen in Medical Applications (CO_2 Laser case study).

A hospital uses a CO_2 laser for surgical procedures such as tumor removal and skin resurfacing. The laser produces a highly coherent and monochromatic beam that allows precise cutting without damaging surrounding tissues.

Questions:

1. Why is population inversion necessary for laser operation?

2. Explain the role of stimulated emission in producing laser light.

3. Why is the CO_2 laser preferred in medical applications?

4. Identify the key characteristics of laser light used in surgery.

Give your own conclusion based on the ~~g~~ questions and the above scenario.

1. Population inversion:

Population inversion is required so that more atoms are in excited state than ground state, enabling laser action.

2. Role of stimulated Emission:

In stimulated Emission, an incoming photon causes an excited atom to emit an identical photon. This produces coherent and intense light.

3. CO_2 Laser is Preferred for:

because.

⇒ Emits Infrared radiation strongly absorbed by tissues.

⇒ Provides precise cutting

⇒ Minimizes bleeding and Damage.

4. Characteristics of Laser light:

⇒ Monochromatic (single wavelenth).

⇒ Coherent (in phase)

⇒ Highly directional.

⇒ High intensity.

Conclusion:

CO_2 lasers are ideal for medical surgery due to their precision, safety and controlled energy delivery, making them effective for delicate procedures.